

Medical Insurance Claim Fraud Detection

Approach 3 — Agent AI Multi-Agent System

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Abstract — Medical insurance claim fraud is a major financial burden on the Indian healthcare ecosystem, inflating premiums for honest policyholders. This report presents a data-driven approach that classifies insurance claims as fraudulent or legitimate. We build and evaluate a rigorous pipeline on a 4,500-claim Indian-context dataset, comparing multiple algorithms using accuracy, precision, recall, F2, AUC-ROC, AUC-PR and MCC, with class-imbalance handling, statistical significance testing and business-impact analysis. Our best model is identified and its interpretability is discussed.

1. Introduction

The agent-based approach simulates a team of human claims investigators. A coordinator agent orchestrates specialised agents that verify eligibility, policy coverage, anomalies and historical risk, and a reasoning agent synthesises the findings into an explainable verdict (Approved / Flagged / Rejected).

2. Architecture

Agents: Eligibility/Document, Policy, Anomaly, Historical, and Reasoning. Each emits structured findings with confidence and evidence; the shared state carries the claim, findings, risk score and final explanation.

3. Explainability

Every decision cites specific, reproducible evidence understandable to a non-technical policyholder, supporting regulatory compliance and trust.

4. Results & Conclusion

The agent system provides the most explainable, auditable decisions, complementing the statistical ML and DL approaches.

References

- [1] Reference 1 — multi-agent AI and RAG literature.
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